Sensitivity Shaping for Latent Modeling

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Abstract: Generative dynamics models enable planning in challenging robotic systems, but safe deployment requires reliably detecting policy-induced out-of-distribution (OOD) transitions. Existing methods typically treat the learned dynamics as fixed and attach post hoc support surrogates. We show that these surrogates can fail when the dynamics are locally insensitive to critical action choices: unsupported control actions may produce latent predictions that resemble demonstrated transitions, suppressing OOD signals despite large true predictive errors. To address this, we introduce support-conditioned control-sensitivity regularization, which promotes sensitive local response to control input changes in learned dynamics in high-support training regions. This preserves control-induced variation while limiting unstable extrapolation due to weak empirical support. Experiments in vision-based obstacle avoidance, manipulation, and real-robot navigation show improved OOD detection and safer closed-loop planning.

Keywords: Generative dynamics model; Out-of-distribution detection

1 Introduction

Generative models that capture system dynamics through latent representations (such as world models [1]) have emerged as promising components for robotics. These models aim to support predictive reasoning beyond reactive control prediction, by mapping high-dimensional observations into a latent manifold that preserves transition-relevant structure. They have enabled planning and control in challenging systems where analytical dynamics are unavailable, including vision-based manipulation [2, 3] and robot navigation under partial observability in dynamic environments [4, 5, 6].

A critical bottleneck of generative dynamics models is that they are vulnerable to deployment-time distribution shift [7, 8]. Such shift arises when the model is queried on out-of-distribution (OOD) state-action inputs, often due to the deviation between deployed policies from the behavior policy which induces changes to the state-visitation distribution [7, 8]. In such OOD cases, bounding prediction errors is challenging. Prior work mitigates this issue by broadening the training distribution, for example through web-scale multimodal data [9, 10, 11, 12, 13] or foundation-model priors [14, 15, 16]. However, there is a consensus on the scarcity of robot interaction data relative to other domains [17, 18, 19, 20], and the scale of the training data alone does not guarantee coverage of task-critical counterfactuals or reliable generalization beyond the training manifold.

These limitations motivate uncertainty-aware analysis that aims to detect transitions that depart from the training support [21]. Existing methods typically design surrogate scores to measure uncertainty, such as: ensembles use inter-model disagreement [22], density models estimate likelihood under a learned data distribution [23], and nearest-neighbor methods measure distance to observed transitions [24]. During planning, these scores are incorporated as penalties to discourage unsupported controls [25]. However, most approaches treat OOD analysis as a post hoc diagnostic, and calibrate surrogates scores with conformal prediction which bounds the false rejection rate for in-distribution inputs at a prescribed level [26]. Yet, safe model-based control also requires the complementary property: genuinely unsupported transitions should be reliably rejected. This property is not guaran-

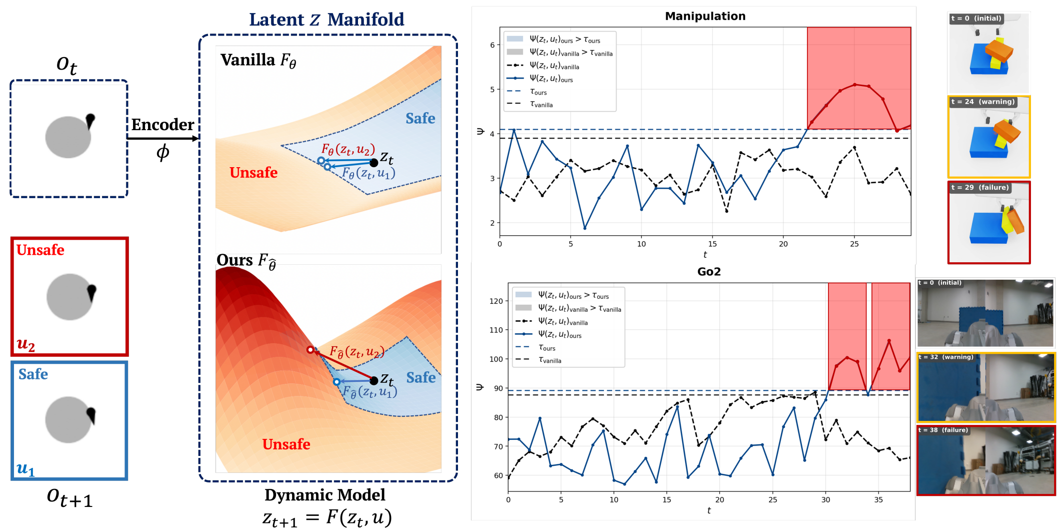

Figure 1: Overview of sensitivity shaping for reliable OOD detection in generative dynamics models. Control-insensitive models respond weakly to control inputs, yielding inaccurate yet in-distribution-looking predictions that obscure OOD detection. **(Left)** At a safety-critical state o_t, different controls lead to distinct outcomes, e.g., a safe transition under u_1 and an unsafe one under u_2. However, the vanilla model trained on failure-free data fail to capture true system consequences due to control-insensitivity which flattens the dynamics landscape over controls, making the predicted next latent states z_1 and z_2 nearly indistinguishable. Our sensitivity regularizer promotes nontrivial control responses, measured via the Frobenius norm of the control-Jacobian within learned dynamics, and improves detectability of unsupported transitions in the latent space. **(Right)** On undemonstrated failure rollouts in manipulation and real-robot navigation, the vanilla model fails to detect OOD before failure, with scores remaining below the calibrated threshold. In contrast, the sensitivity-regularized model flags OOD earlier, with red shading marking steps classified as OOD.

teed by calibration alone; it depends on whether the surrogate score captures true support mismatch along rollout distributions and thereby provides an informative proxy for prediction error. Importantly, surrogate-measured support mismatch need not align with actual prediction error (Figure 1).

In this paper, we improve OOD analysis for generative dynamics models by shaping the local sensitivity in the learned dynamics landscape. We identify *control insensitivity* as a critical failure mode, where learned dynamics respond too weakly to control inputs and underrepresent the true responses. Under this failure mode, undemonstrated controls yield predictions that spuriously resemble the training distribution despite large true predictive errors, causing standard support-based OOD surrogates to fail. To address this, we introduce a sensitivity regularizer applied to well-supported regions of the training distribution. The regularizer encourages nontrivial Frobenius norms on the control Jacobian, encouraging locally responsive dynamics without attempting to guarantee accurate prediction for arbitrary unseen controls. By preventing control-induced predictions from collapsing to overly similar latent transitions, the method makes unsupported transitions easier to detect.

Overall, we make the following contributions. First, we analyze control sensitivity in generative dynamics models through a vision-based obstacle-avoidance case study, demonstrating how control insensitivity degrades OOD detection (Section 3). Second, we propose a support-conditioned sensitivity regularizer that promotes local responsiveness of the learned dynamics and improves detectability of unsupported transitions (Section 4). Finally, we show that this method enhances control sensitivity, improves OOD detection, and enables more reliable OOD-aware planning across simulated and real-robot experiments (Section 5).

Related Work. Generative dynamics models learn latent representations and transition dynamics from high-dimensional observations for prediction, planning, and control [1]. Recurrent State-Space Models (RSSMs) [27], trained with variational objectives [28, 29, 30], form the basis of the Dreamer family [31, 32, 33, 34], which optimizes policies through imagined latent rollouts. These models have been applied to embodied robotic tasks such as manipulation, navigation, and locomotion [2, 4, 35, 5, 36]. However, learned dynamics can become unreliable under deployment-time distribution shift [37, 38], which is especially problematic for safety-critical control [39, 40, 41]. Existing out-of-distribution (OOD) analysis methods monitor such failures using distance-based scores [42, 43, 44, 45], density- or energy-based criteria [46, 47, 48, 49], or ensemble uncertainty [8, 38, 7], often to trigger runtime interventions or safety filters [50, 51, 52, 25]. These approaches are largely post hoc and depend on the learned representation. In contrast, we show that OOD detectability can be limited by the learned latent dynamics themselves, and propose to improve it by shaping control sensitivity during dynamics learning.

2 Preliminaries

Generative Dynamics Models. Consider a discrete-time dynamical system. At each time step t, the agent observes $o_t \in \mathcal{O}$, applies $u_t \in \mathcal{U}$, and transitions to $o_{t+1} = f(o_t, u_t)$ where $f : \mathcal{O} \times \mathcal{U} \to \mathcal{O}$. Generative dynamics models approximate f in a lower-dimensional latent space $\mathcal{Z}$ involving

$$\text{Encoder: } z_t \in \mathcal{Z} \sim \phi_\theta(\cdot \mid o_t), \qquad \text{Dynamics: } \hat{z}_{t+1} \in \mathcal{Z} \sim F_\theta(\cdot \mid z_t, u_t). \tag{1}$$

The key training objective is latent predictive consistency: the dynamics prior predicted from (z_t, u_t) should align with the encoder posterior of the next observation, i.e, $F_\theta(\cdot \mid z_t, u_t) \approx \phi_\theta(\cdot \mid o_{t+1})$.

Out-of-Distribution (OOD) Analysis. OOD analysis studies whether a learned model is queried outside the support of its training data. We use an OOD surrogate: $\Psi : \mathcal{Z} \times \mathcal{U} \to \mathbb{R}^+$, where larger values indicate lower support. Common choices include: k *nearest neighbor* (kNN) distances to training latent transitions [24], *Flow Matching* negative log-density estimated via a learned flow to a Gaussian base [23], and *Ensemble* disagreement across independently dynamics models [22]. Inductive conformal prediction (ICP) [26] provides a principal way to convert a chosen surrogate into an OOD boundary: on a held-out in-distribution calibration set, for a prescribed level α, τ_α is chosen as the the empirical $(1-\alpha)$-quantile of Ψ. A pair (z_t, u_t) is classified as OOD if $\Psi(z_t, u_t) > \tau_\alpha$, yielding a false rejection probability for in-distribution samples bounded by α.

3 Analysis: Control-Conditioned Sensitivity in Latent Dynamics

Control insensitivity creates a fundamental failure mode for OOD analysis: when the learned dynamics depend weakly on controls, they may produce inaccurate predictions that nevertheless appear compatible with the training distribution. Since OOD surrogates measure support rather than transition correctness, such errors can evade detection. We study this issue in safety-critical systems [53], where failure-free data often provide limited control diversity near critical regions, making control-dependent transitions weakly identifiable and inducing local control insensitivity.

3.1 Running Example: Vision-Based Dubins Obstacle Avoidance

We illustrate this failure mode in a vision-based obstacle-avoidance case study. The agent follows constant-speed Dubins dynamics with a scalar steering-rate input and observes bird's-eye-view images. We train generative dynamics models using DreamerV3 [33] on collision-free trajectories collected with expert and random controls. Figure 2-Left shows that the trained model can accurately predict future states for demonstrated state-control pairs.

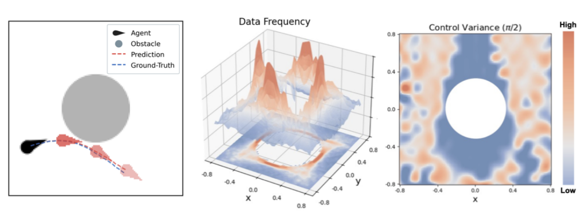

Figure 2: Dubins system. **(Left)** RGB obesrvations. Overlays show that model-predicted future states accurately tracks the ground-truth. **(Middle)** Training-data density over the spatial map, showing denser coverage near the obstacle and sparser coverage farther away. **(Right)** Spatial control-variance map for heading $\pi/2$. Low variance near the obstacle indicates limited control diversity under safe constraints.

Training Data Analysis. Figure 2-Middle shows that samples concentrate near the central obstacle, while regions farther away are comparatively sparse. In contrast, the empirical

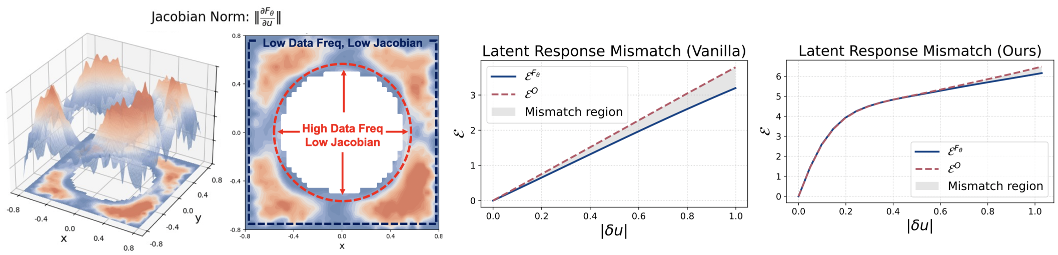

Figure 3: Sensitivity analysis in the obstacle-avoidance system. From left to right: **(a,b)** Control-Jacobian norms for demonstrated latent state-control pairs, project onto the 2D position space and averaged within discretized bins; blue indicates lower sensitivity and red indicates higher sensitivity. Low sensitivity regions arise from limited control diversity or sparse data coverage. **(c)** The vanilla, control-insensitive dynamics deviate from the true system responses even under small perturbations. **(d)** The sensitivity-regularized dynamics better match the true response, substantially reducing small-perturbation mismatch.

control variance is low near the obstacle (Figure 2-Right), since the exclusion of failure trajectories restricts near-boundary demonstrations to controls that remain safe. This reduced control diversity weakens the identifiability of control-dependent dynamics in safety-critical regions.

3.1.1 Control Sensitivity of the Learned Dynamics

Measuring Control Sensitivity. We first formalize the sensitivity measure. For an arbitrary latent state-control pair (z_t, u_t), we fix z_t and view the learned transition as a differential function of the control input: $F_\theta(z_t, \cdot) : \mathcal{U} \subseteq \mathbb{R}^{|\mathcal{U}|} \to \mathcal{Z} \subseteq \mathbb{R}^{|\mathcal{Z}|}$. Its Jacobian $J_u F_\theta(z_t, u_t) \in \mathbb{R}^{|\mathcal{Z}|\times|\mathcal{U}|}$ maps a control perturbation direction δu to the first-order change in the predicted next latent state:

$$\mathrm{d}F_\theta(z_t, u_t)[\delta u] = J_u F_\theta(z_t, u_t) \cdot \delta u := \frac{\mathrm{d}}{\mathrm{d}\varepsilon} F_\theta(z_t, u_t + \varepsilon\, \partial u)\bigg|_{\varepsilon=0}. \tag{2}$$

We measure control sensitivity by the Frobenius norm of this Jacobian, i.e., $\|J_u F_\theta(z_t, u_t)\|_F = (\sum_{i=1}^{|\mathcal{Z}|} \sum_{j=1}^{|\mathcal{U}|} (J_u F_\theta(z_t, u_t))_{ij}^2)^{1/2}$. This norm aggregates first-order responses across control directions, serving as a continuous analogue of average sensitivity for Boolean functions [54], and a standard measure of neural network sensitivity [55].

Sensitivity Results. Figure 3-(a,b) visualizes control-Jacobian Frobenius norms over demonstrated state-control pairs. The results shows two low-sensitivity regimes: sparse outer regions near the position-space boundary, and near-obstacle regions with substantial state coverage but limited control diversity. Low sensitivity indicates that the learned dynamics respond weakly to changes in the control input, producing similar next-latent predictions for distinct controls. However, the true Dubins dynamics (see Appendix) demand stronger sensitivity: the steering-rate input affects the heading with state-independent gain, while constant forward speed induces position changes of fixed magnitude. Thus, for a fixed control perturbation, the true response should be comparable across states and should not vanish near the safety boundary.

To quantify this mismatch, we compare the model-predicted dynamics response with the true system response. For an arbitrary latent-valued transition map H, define its control-induced response as $\Delta_H(x, u; \partial u) \triangleq H(x, u + \delta u) - H(x, u)$. This measures the change in H at input x induced by perturbing the control from u to $u + \partial u$. Let $\mathbb{G} : \mathcal{O} \times \mathcal{U} \to \mathcal{O}$ denote the one-step observation transition, and write $H_O(o, u) \triangleq \phi_\theta(\mathbb{G}(o, u))$. For (o_t, u_t), with $z_t \sim \phi_\theta(\cdot \mid o_t)$, we measure

$$\mathcal{E}^{F_\theta}(z_t, u_t;\, \delta u) \triangleq \|\Delta_{F_\theta}(z_t, u_t;\, \partial u)\|_2^2, \quad \mathcal{E}^{O}(o_t, u_t;\, \delta u) \triangleq \|\Delta_{H_O}(o_t, u_t;\, \partial u)\|_2^2, \tag{3}$$

where $\mathcal{E}^{F_\theta}$ is the model-predicted response and $\mathcal{E}^O$ is the true response in the same latent space. Figure 3-c shows that, over near-obstacle low-sensitivity samples identified in Figure 2, the gap between $\mathcal{E}^{F_\theta}$ and $\mathcal{E}^O$ grows with $\|\delta u\|$, indicating that learned dynamics increasingly underestimate control-induced variation away from demonstrated controls. Importantly, since the gap appears even for small perturbations, the mismatch is local and can compound rapidly under recursive rollouts, showing that the learned control-insensitive dynamics underrepresents the true system dynamics.

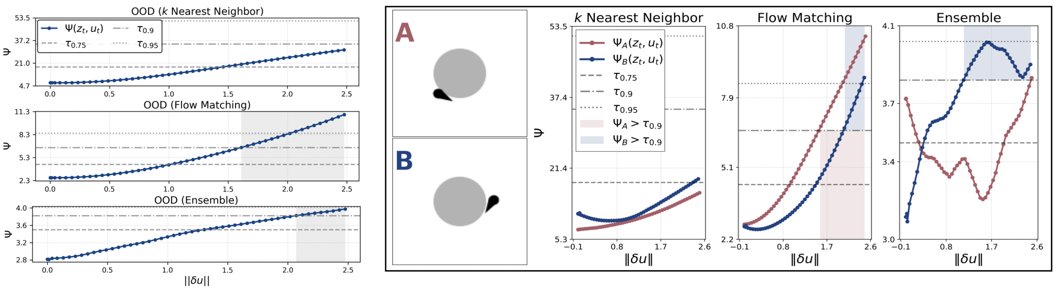

Figure 4: OOD surrogate responses to perturbations. **(Left)** Average surrogate scores over near-obstacle samples as demonstrated controls are perturbed. Dashed horizontal lines denote conformal thresholds at coverage levels 0.75, 0.90, and 0.95; shaded regions indicate perturbations classified as OOD under the 0.90-coverage threshold. Although scores generally increase with $\|\delta u\|$, each misses some safety-critical perturbations to varying degrees. **(Right)** Qualitative sample-level responses. In Scenario A, the ensemble score decreases under perturbations because shared control-insensitive dynamics cause disagreement to overlook transition error.

3.1.2 Impact on OOD Analysis

We next discuss how control insensitivity degrades OOD analysis. The failures stem from a mismatch between latent transition geometry and prediction error: if perturbed controls yield predicted transitions close to the demonstrated distribution, support-based scores may not reflect the true transition error as these surrogates mainly assess compatibility with the training distribution.

Empirical Results. Over near-obstacle, low-sensitivity samples, we perturb demonstrated controls by δu and measure scores as $\|\delta u\|$ increases; Figure 4 reports results, showing that OOD scores need not track increasing control deviation. For *kNN* which measures latent distance to training samples, control-insensitive predictions can falsely resemble training distribution and therefore receive small nearest-neighbor distances. Accordingly, its average scores remain below the 0.90-coverage OOD threshold across the full range of $\|\delta u\|$. *Ensemble* scores appear more responsive on average but can be non-monotonic at the sample level: when members share the same control-insensitive bias, disagreement reflects inter-model variance rather than shared transition error and may even decrease with $\|\delta u\|$. *Flow matching* performs best, but perturbed transitions near the training manifold may still be mapped through well-trained regions of the flow and assigned high likelihood. Iterative flow integration can further smooth small deviations, reducing the contrast between perturbed and demonstrated transitions and causing some safety-critical perturbations to be missed (Figure 5). These results show that post hoc OOD surrogates cannot fully compensate for collapsed control-conditioned transitions.

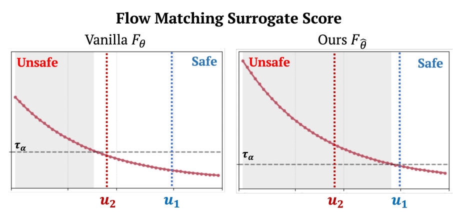

Figure 5: Flow-matching scores for the state o_t in Figure 1. Dashed lines indicate calibrated thresholds. For the unsafe control u_2, vanilla dynamics **(left)** assigns an in-distribution score due to control-insensitivity, whereas the proposed sensitivity-regularized dynamics **(right)** correctly identifies the transition as OOD.

4 Method: Support-Conditioned Control Sensitivity Regularization

We mitigate control insensitivity to facilitate OOD detection by regularizing the learned dynamics to maintain nontrivial control-Jacobian norms in well-supported regions of the training distribution. The goal is not to guarantee accurate prediction under arbitrary controls, but to prevent undemonstrated controls from collapsing to overly similar latent transitions. For a latent state–control pair (z_t, u_t) and valid control perturbations δu, a first-order expansion shows that the average local response to control perturbations is governed by the control Jacobian:

$$\mathbb{E}_{\delta u}\left[\left\|\Delta_{F_\theta}(z_t, u_t; \partial u)\right\|_2^2\right] \approx \mathbb{E}_{\delta u}\left[\|J_u F_\theta(z_t, u_t)\, \delta u\|_2^2\right] = \frac{\bar{\sigma}_u^2}{|\mathcal{U}|}\, \|J_u F_\theta(z_t, u_t)\|_F^2, \tag{4}$$

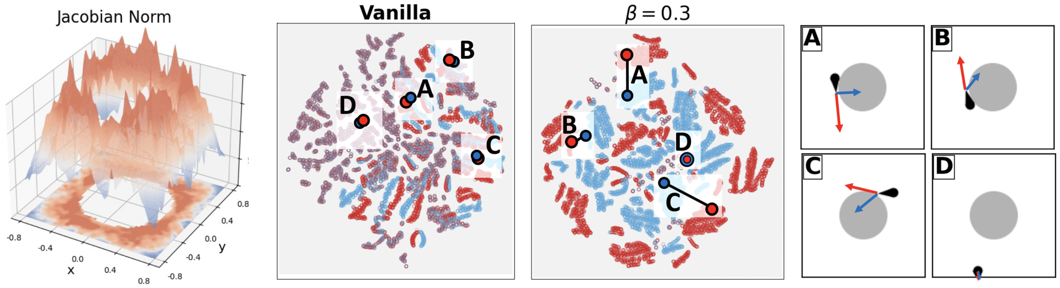

Figure 6: Sensitivity analysis with sensitivity-regularized dynamics. From left to right: **(a)** Control-Jacobian norms of the regularized dynamics, showing increased responsiveness in near-obstacle regions that were previously control-insensitive. **(b,c)** PCA+t-SNE visualizations of latent transitions under demonstrated controls (red) and undemonstrated controls (blue). The vanilla model (b) exhibits substantial overlap, indicating under-represented control-induced variation, wheras the regularized model (c) shows improved separation achieved by the proposed non-contrastive sensitive regularization. **(d)** Annotated states for marked points in (b,c).

where $\bar{\sigma}_u = \mathbb{E}_{\partial u}\left[\|\partial u\|_2\right]$ is the expected perturbation magnitude. While the Frobenius norm does not lower-bound every perturbation direction, enforcing a non-vanishing norm promotes aggregate control responsiveness and mitigate collapse onto demonstrated transitions.

Support-Conditioned Targeting. Uniformly enforcing strong local responsiveness across the training manifold can degrade prediction in weakly supported regions: a global Jacobian constraint may increase local Lipschitz constants and amplify extrapolation errors through sensitive transitions [56]. We therefore apply sensitivity regularization only in high-supported regions, where empirical observations sufficiently constrain the latent dynamics. To identify these regions, we use kNN surrogate on in-distribution samples as a parameter-free proxy for local support. Since this assessment ranks demonstrated inputs, it is not subject to the control-insensivity artifacts.

Control Sensitivity Regularization. Since the encoder evolves during training, we periodically update the high-support set, denoted as $\bar{\mathbb{D}}$, by retaining the β-fraction of samples with the smallest latent-space kNN distances. On this set, we promote local control sensitivity with

$$\mathcal{L}_{\text{reg}} = \mathbb{E}_{\substack{(o_t,u_t)\sim\bar{\mathbb{D}} \\ z_t\sim\phi_\theta(\cdot|o_t)}} \left[\left(g - \left\| \frac{\partial F_\theta(z_t,u_t)}{\partial u_t} \right\|_F^2 \right)_+ \right], \tag{5}$$

where $(\cdot)_+ = \max(\cdot, 0)$ and $g > 0$ sets the lower bound on the squared norm, penalizing vanishing sensitivity while avoiding unbounded growth. To reduce computational cost, we estimate the Frobenius norm using Hutchinson's identity [57]. The final objective is: $\mathcal{L} = \mathcal{L}_{\text{dyn}} + \lambda_{reg}\mathcal{L}_{\text{reg}}$, where $\mathcal{L}_{\text{dyn}}$ denotes the standard generative dynamics training loss and λ_{reg} is the regularization weight.

5 Experiments

5.1 Dubins Obstacle Avoidance

We first evaluate the proposed method on the Dubins system introduced in Section 3; training configurations are provided in the Appendix. We evaluate whether the regularized dynamics recover stronger control sensitivity and improve downstream OOD-aware planning.

Sensitivity Analysis. Figure 6-a shows that, compared to the unregularized model (Figure 3), the proposed sensitivity-regularized dynamics exhibits substantially stronger responsiveness in near-obstacle regions. To inspect the induced latent geometry, Figure 6-(b,c) visualizes predicted next-latent states for the top 20% high-support training samples. For each sample, we construct an undemonstrated transition by replacing its control with the one farthest from the empirical control distribution. Given the pairs (z_t, u_t), we unroll the learned dynamics and and project the predicted next-latent states $z_{t+1} \sim F_\theta(\cdot \mid z_t, u_t)$ using PCA followed by t-SNE. For the unregularized model, demonstrated and undemonstrated transitions substantially overlap, indicating that control-insensitive dynamics map distinct controls to overly similar latent predictions. In contrast, the regularized model yields improved separation despite using a *non-contrastive* Jacobian regularizer. The remaining overlap typically corresponds to common near-initial or near-terminal states. Figure 3-d shows that the regularized model more closely tracks the ground-truth latent response $\mathcal{E}^O$.

OOD-Aware Planning. We use calibrated OOD thresholds as one-step safety filters for a random reference controller. At each time step, we reject the proposed control u_t^{ref} if its induced transition has an OOD score above the calibrated threshold. Upon rejection, we execute the sampled alternative with the lowest OOD score. We report: ρ_{succ}, the fraction of safe trajectories; ρ_{filt}, the fraction of rejected reference control; and ρ_{viol}, the fraction of executed transitions classified as OOD.

Method	Ψ	$\rho_{\mathrm{succ}} \uparrow$	ρ_{filt}	$\rho_{\mathrm{viol}} \downarrow$
Vanilla	kNN	0.600	0.340	0.273
	FM	0.910	0.309	0.153
	ENS	0.380	0.459	0.327
Ours	kNN	0.800	0.323	0.166
	FM	**0.960**	0.323	**0.104**
	ENS	0.440	0.425	0.297

Table 1: Obstacle-avoidance with filtering of a random policy (200 trials).

Results. Table 1 compares safety filters built from vanilla and the proposed regularized dynamics. Flow matching outperforms kNN, consistent with the preceding analysis showing that flow matching is most responsive. Ensemble consistently underperforms, likely due to spurious inter-member agreement on far-OOD inputs [45, 24]; this issue is amplified in the one-step filtering where the surrogate must provide reliable local rankings of candidate controls. Across all three surrogates, filters based on the regularized dynamics improve over their vanilla counterparts, achieving higher success rates and lower OOD violation rates. These results indicate that enhanced control sensitivity improves OOD-aware safety filtering by increasing detectability of unsupported candidate controls.

5.2 Block Plucking Manipulation

Setup. We next consider a manipulation task in which a Franka arm extracts the middle block from a three-block stack without dropping the top block (Figure 7). Observations are RGB images from wrist-mounted and tabletop cameras, together with 7-dimensional proprioception. The controls are a 6-DoF end-effector pose increment and a discrete gripper command.

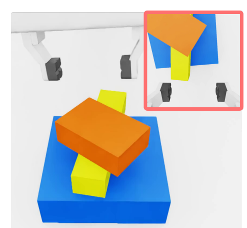

Figure 7: Observations in block-plucking.

We evaluate OOD-aware planning using the ensemble surrogate and a reachability-based safety filter [25]. Unlike the one-step filter, this filter is trained on imagined latent rollouts to reason about long-horizon OOD risk. Given an OOD surrogate Ψ and a calibrated threshold τ_α at calibration level α, the reachability filter learns to avoid trajectories that enter latent states z from which all admissible controls are OOD, i.e., $\min_{u \in \mathcal{U}} \Psi(z, u) > \tau_\alpha$. Because trajectories may remain safe but fail to complete the task, we replace success rate with the failure rate ρ_{fail} and the incompletion rate ρ_{inc}. We also report the reconstruction error $\mathcal{E}_{\mathrm{recon}}$, defined as the negative log-likelihood of the observations under the trained decoder's reconstruction distribution, as a proxy for how closely executed trajectories remain within the training support.

Results. Table 2 evaluates a mediocre reference policy with $\rho_{\mathrm{fail}} \approx 0.25$. Compared with the filter trained on vanilla dynamics, the filter trained on sensitivity-regularized dynamics achieves a lower failure rate with moderate intervention. It also produces the lowest reconstruction error, suggesting that the filtered trajectories remain the closest to the training support.

Table 3 filters a policy trained on imagined rollouts to maximize the learned reward [33]. This setting evaluates the full offline pipeline: learning dynamics and rewards from data, then training both policy and OOD filter entirely through latent imagination. We note that larger α tightens the OOD threshold, yielding a more conservative filter: $\alpha = 0.2$ minimizes reconstruction error and failures, but maximizes task incompletions by suppressing progressive controls. Conversely, a smaller α improves in-distribution coverage but enables aggressive, risker planning when the learned dynamics or OOD surrogates are imperfect. Notably, the filter built from sensitivity-regularized dynamics at $\alpha = 0.05$ outperforms the vanilla-dynamics filter at $\alpha = 0.1$, suggesting that improved control sensitivity preserve safety while reducing the need for overly conservative calibration.

Filter	α	$\rho_{\text{fail}} \downarrow$	ρ_{inc}	ρ_{filt}	$\mathcal{E}_{\text{recon}} \downarrow$
None	N/A	0.243	0.096	0.000	2335.99
Vanilla	0.10	0.196	0.047	0.242	2215.32
Ours	0.10	**0.150**	0.047	0.269	**2196.11**

Table 2: Block-plucking with filtering of a mediocre policy (100 trials).

Filter	α	$\rho_{\text{fail}} \downarrow$	ρ_{inc}	ρ_{filt}	$\mathcal{E}_{\text{recon}} \downarrow$
Vanilla	0.10	0.118	0.013	0.154	2207.55
Ours	0.20	**0.046**	0.157	0.322	**2036.50**
Ours	0.10	0.047	0.053	0.199	2146.66
Ours	0.05	0.085	**0.007**	0.100	2204.37

Table 3: Block-plucking with filtering of the policy trained from the same dynamics model (100 trials).

5.3 Real-Robot Static-Obstacle Avoidance

Setup. Lastly, we study a real-robot obstacle-avoidance system using images from two body-mounted cameras (Figure 8). We collect 60 minutes of collision-free teleoperation using three discrete commands: turn left, move straight, and turn right, all executed at a fixed forward speed of $0.5\,\mathrm{m/s}$. Despite the discrete control space, the control-sensitivity view still applies, as control insensitivity is data-induced limitation of learned dynamics and discrete commands are embedded as continuous inputs to the learned dynamics.

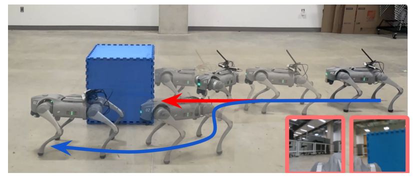

Figure 8: Real-robot obstacle avoidance system. The unfiltered trajectory (red) results in failure, whereas the flow-matching filter (blue) avoids it. Bottom-right insets show body-camera observations.

Method	Ψ	$\rho_{\text{succ}} \uparrow$	ρ_{filt}	$\rho_{\text{viol}} \downarrow$
Vanilla	kNN	0.240	0.047	0.027
	FM	0.900	0.343	0.155
	ENS	0.150	0.543	0.507
Ours	kNN	0.380	0.040	**0.026**
	FM	**1.000**	0.326	0.154
	ENS	0.750	0.592	0.557

Table 4: Real-robot with filtering of a random policy (50 trials).

Results. Table 4 reports filtering results for a random reference policy, using one-step filters for kNN and flow matching and a reachability-based filter for ensemble. The kNN surrogate provides limited performance, reflecting the difficulty of using latent distances in real-world visual settings where covering all observation variations is impossible. Flow matching performs best under the available data budget and improves further when paired with the sensitivity-regularized dynamics, suggesting that parametric density surrogates can effectively model high-dimensional support without exhaustive coverage. The ensemble filter incurs frequent interventions because long-horizon imagined rollouts can amplify spurious OOD classification. Nevertheless, its large ρ_{succ} improvement with regularized dynamics indicates that reachability-based filtering is particularly sensitive to control-conditioned transition collapse. We interpret ρ_{viol} cautiously, as unmodeled camera variations can induce observation-level OOD events orthogonal to our focus. Overall, we show that that enhancing control sensitivity improves OOD detection and OOD-aware planning on hardware. Supplementary videos provide comparisons of the resulting behavioral differences.

6 Conclusion

This work identifies control insensitivity in generative dynamics models as a key failure mode for OOD analysis. We show that weak sensitivity can cause unsupported controls to produce in-distribution-looking predictions, systematically suppressing OOD signals. To mitigate this, we introduce a support-conditioned sensitivity regularizer that enforces non-vanishing control-Jacobian norms in high-support regions of the training distribution. Through sensitivity analyses and safety-filtering experiments, we demonstrate improved OOD detection and safer model-based control.

Limitations. The proposed regularizer relies on latent kNN-based support estimates; poor representations or distance metric may lead to misidentified support regions. Periodic support-set updates also add computational cost for large datasets. Future work should study scalable and adaptive support estimation, richer visual representations, and broader hardware validation.

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